DATABASE SYSTEMS — COMPREHENSIVE EXAM STUDY NOTES

Course Code & Title: CS Database Systems | Target: Midterm & Final Examinations

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SECTION 1: FUNDAMENTALS & CORE DEFINITIONS

Data: Raw, unorganized, and unprocessed facts, numbers, text, images, or symbols that lack context or specific meaning on their own (e.g., numbers like "40", "1024").

Information: Data that has been processed, organized, structured, or contextualized so that it becomes meaningful and useful for decision-making (e.g., "Score: 40/50" or "Temperature: 40°C").

Database: An organized collection of logically related data stored electronically in a structured manner so that it can be easily accessed, managed, updated, and queried.

Database Management System (DBMS): A software layer that acts as an intermediary between end users/applications and the underlying physical database. It provides tools to define, create, maintain, manipulate, and control access to data.

Common Real-World Database Examples

- **Bank Accounts:** Tracks deposits, withdrawals, transfers, and customer information.
- **Web Indexes & Search Engines:** Stores pages, links, and keywords for quick searching.
- **Library Catalogues:** Tracks books, authors, availability, and borrower records.
- **Medical & Hospital Records:** Manages patient history, prescriptions, and doctor appointments.
- **Stock Control & Inventory:** Manages product counts, supply chains, and order processing.

SECTION 2: DATABASE SYSTEM COMPONENTS & USERS

1. Four Major Components of a Database System

COMPONENT	DESCRIPTION & FUNCTION
Data	The actual operational information stored in the database, along with metadata (description of the data structure).
Software	Includes the DBMS software itself, application software, utilities, and operating system interfaces.
Hardware	Physical infrastructure including hard drives, servers, memory (RAM), and networking cables/devices.
Users	The people who interact with the system (administrators, developers, and end users).

2. Categories of Database Users

- **End Users:** People who use the database system to perform daily tasks (e.g., bank tellers, retail cashiers, airline booking agents). They interact via user interfaces.
- **Application Developers:** Software programmers who write applications using languages like C++, Java, or Python to interface with the database.
- **Database Administrator (DBA):** The person responsible for overall management, security, performance monitoring, physical design, backup, and user authorization of the database.
- **Database Systems Programmer:** Computer engineers who build, write, and optimize the internal code of the DBMS software itself (e.g., writing MySQL engine code).

EXAM TIP — QUESTION HANDLING: If a question asks, *"What is the difference between a Data Administrator (DA) and a Database Administrator (DBA)?"* Answer: A **DATA ADMINISTRATOR (DA)** handles corporate policy, data strategy, and business planning (often high-level management/CIO level). A **DATABASE ADMINISTRATOR (DBA)** focuses on

SECTION 3: FILE-BASED SYSTEMS VS. DBMS

1. Traditional File-Based Systems

Before DBMS existed, organizations stored data in manual paper files or simple computer files (like Excel or plain text files) where each department maintained its own separate files.

2. Major Limitations of Traditional File-Based Systems

- **Data Duplication / Redundancy:** The same piece of information is stored in multiple files, wasting storage space (e.g., student address saved separately in Accounts and Library files).
- **Data Inconsistency:** When duplicated data is updated in one file but not in another, leading to conflicting records.
- **Data Dependence:** File structure is tightly coupled with program code. If file format changes, every application program must be rewritten.
- **No Standardized Ad-Hoc Querying:** Users cannot ask quick, custom, on-the-spot questions; complex custom code must be written for every new report.
- **Lack of Security, Recovery, and Concurrency Controls:** No built-in protections against simultaneous user conflicts, data corruption, or unauthorized access.

3. Comparison: Traditional File System vs. Database Management System

FEATURE	TRADITIONAL FILE SYSTEM	DATABASE MANAGEMENT SYSTEM (DBMS)
Data Redundancy	High redundancy; data duplicated across departments.	Minimized redundancy; central repository shared by all.
Data Consistency	Low; updating one file leaves others outdated.	High; updating central data reflects everywhere instantly.
Data Independence	No independence; programs depend on file structures.	High independence; changes in data structure don't break code.
Security & Access	Weak security; hard to enforce fine permissions.	Robust security; roles, privileges, and encryption used.
Concurrent Access	Poor; multiple users modifying files causes overwrite errors.	Strong; multi-user concurrency controlled via locking mechanisms.
Backup & Recovery	Manual and error-prone.	Automated tools for logging, crash recovery, and backup.

SECTION 4: CORE TERMINOLOGY & DATA HIERARCHY

Enterprise: The organization or business entity using the system (e.g., University, Bank, Hospital).

Repository: A centralized physical location where all metadata and data definition files are stored.

Entity: A real-world person, place, thing, event, or concept about which data is collected (e.g., `Student`, `Course`, `Account`).

Data Dictionary: A repository file containing descriptions of table schemas, column names, rules, and data structures.

Attribute: A characteristic or property describing an entity (e.g., `Student_Name`, `Age`, `GPA`). Also called a field or domain.

Metadata: "Data about data." It details attribute names, data types (e.g., INTEGER, VARCHAR), size limits, formats, and constraint rules.

Key Types & Identification

- **Key / Candidate Key:** An attribute or set of attributes used to identify or locate specific records in a table.
- **Primary Key (PK):** A specific attribute or minimal combination of attributes that **uniquely identifies** each record/row in a table (e.g., `Student_ID`). It cannot contain duplicate or null values.

SECTION 5: BASIC CRUD OPERATIONS & DATABASE ENVIRONMENT

1. The 4 Fundamental Database Activities (CRUD)

Every database system application performs four primary operational activities on data:

- **Create (Insert):** Adding new records or entities into the database (e.g., registering a new student).
- **Read (Retrieve):** Querying, searching, and viewing existing data without modifying it (e.g., printing a transcript).
- **Update (Modify):** Changing existing data values in the database (e.g., updating a phone number or grade).
- **Delete (Remove):** Permanently removing outdated or invalid records from the database (e.g., deleting a canceled course).

2. Environmental Architecture of a DBMS

A complete database system software environment integrates two distinct execution phases:

Design-Time Tools:
- Table Creation Tools
- Form Builder / Generator
- Query Builder (SQL Editor)
- Report Designer
- Procedural Language Compiler (4GL)

Run-Time Processor Tools:
- Run-Time Form Processor
- Query Engine / Optimizer
- Report Execution Engine
- User Application Interfaces
- Database Kernel Engine

SECTION 6: DATABASE CLASSIFICATIONS & SYSTEM ARCHITECTURES

1. Relational Databases (RDBMS) vs. NoSQL Databases

CATEGORY	RELATIONAL DATABASE (SQL / RDBMS)	NOSQL DATABASE
Structure	Organized into structured tables with rows (records) and columns (fields).	Non-tabular (Document, Key-Value, Wide-Column, Graph).
Schema	Strict, pre-defined schema required before inserting data.	Dynamic, flexible, or schema-less structure.
Guarantees	Strict ACID compliance (Atomicity, Consistency, Isolation, Durability).	BASE compliance (Basically Available, Soft-state, Eventual consistency).
Best For	Structured data, financial transactions, complex relational queries.	Unstructured big data, real-time analytics, flexible web apps.
Examples	MySQL, Oracle Database, PostgreSQL, Microsoft SQL Server.	MongoDB (Document), Redis (Key-Value), Cassandra, Neo4j.

2. Physical Hardware Architecture Classification

- **PC / Single-User Database:** Designed for individual users on a personal computer (e.g., MS Access, SQLite). Simple and limited scale.
- **Centralized Database System:** All data, the DBMS software, and storage reside on a single central computer or main mainframe server accessible over a network (e.g., a university server storing all student records).
- **Decentralized System:** A collection of multiple independent, stand-alone databases located on separate, non-networked computers. No central sharing occurs (e.g., each academic department keeping isolated spreadsheet files).
- **Client/Server Database:** The database resides on a dedicated server while client machines run application software that sends SQL requests to the server over a local or wide network.
- **Distributed Database System (DDB):** A collection of logically interrelated databases distributed across multiple physically separate locations connected by a computer network (e.g., global bank branches synchronized across continents).

SECTION 7: HISTORICAL CONTEXT & INDUSTRY LANDSCAPE

1. Brief History of Databases

- **1950s – 1960s:** Application systems were custom-built for specific organizational needs using primitive file-based storage. High redundancy and high maintenance.
- **1970: E.F. Codd** published his landmark paper introducing the **Relational Model**, revolutionizing database management with mathematical relational algebra.
- **1979:** Ashton-Tate released the first commercial microcomputer DBMS (dBase), bringing databases to desktop computers.

2. Industry Standard ACID Properties (Critical RDBMS Rules)

A — Atomicity: "All or nothing." Either every operation in a database transaction completes successfully, or the entire transaction is aborted and rolled back.

C — Consistency: Transactions must take the database from one valid state to another, maintaining all schema constraints and rules.

I — Isolation: Concurrent transactions execute independently without interfering with one another or reading partial states.

D — Durability: Once a transaction is committed, its changes are permanently saved in memory/disk, even during system failure or power loss.

SECTION 8: EXAM PREPARATION STRATEGY & EXPECTED QUESTIONS

STRATEGY FOR LONG/SHORT QUESTIONS: Teachers evaluate exam papers based on accuracy, structure, and proper usage of terminology. When answering questions on this material:

1. Always define terms using formal definitions first, then provide a simple 1-sentence explanation.
2. Always provide real-world examples (e.g., mention bank accounts or university registration).
3. Use structured bullet points or comparison tables rather than long, block text paragraphs.

Predicted Exam Questions & Sample Answers

Q1: "Why is a database management system (DBMS) preferred over a traditional file-processing system? List four reasons."

Sample Answer Structure: A DBMS is preferred because: (1) It controls data redundancy through a single shared repository, (2) It ensures data consistency by eliminating update anomalies, (3) It provides data independence, separating programs from storage formats, and (4) It enforces security privileges and concurrent multi-user access control.

Q2: "What is Metadata and where is it stored?"

Sample Answer Structure: Metadata is "data about data." It defines the attributes, data types, field lengths, permissions, and structural constraints of data elements. In a DBMS, metadata is stored in the **Data Dictionary** inside the database repository.

Q3: "Differentiate between Centralized, Decentralized, and Distributed Database Systems."

Sample Answer Structure:
• **Centralized:** All data resides on one server connected to a network.
• **Decentralized:** Stand-alone, non-networked databases stored on separate systems.
• **Distributed:** Logically connected database fragments physically spread across multiple networked computers across locations.

***** END OF EXAM STUDY NOTES — GOOD LUCK WITH YOUR PREPARATION *****

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